

DECISION JOURNAL

Week 7 Model Choice — CariSurg MedTech Pathways 2026

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1 Context

- **Baseline performance:** The Week 6 Logistic Regression baseline achieved ESI-1 Recall of 0.500, correctly identifying 8 of 16 critically ill patients in the held-out test set. The ED Board asked: what does a more sophisticated model buy us, and is it worth the extra complexity?
- **The ESI-1 imbalance problem:** ESI 1 patients represent only 0.14% of the dataset (77 of 55,121 encounters). Every complex model trained with default settings predicted zero ESI 1 patients, the extreme imbalance made the cost of missing them negligible relative to 44,000 majority-class patients. The breakthrough was threshold tuning: lowering the ESI-1 probability decision boundary to half the mean ESI-1 probability observed among true ESI 1 training patients.

2 Alternatives Considered

Model	Key Strength	Key Weakness	IT Governance Risk
Gradient Boosting + Threshold	Strongest on tabular clinical data (Chen et al., 2023)	Low interpretability; requires SHAP for any per-patient explanation; ~4 min training	Medium-High
MLP Neural Network + Threshold	Most flexible feature combinations	Lowest interpretability of all three; rarely outperforms tree models on tabular data; hardest to audit for demographic fairness	High
Random Forest + Threshold ✓ CHOSEN	Medium interpretability via feature importances; class_weight=balanced; 35s training	Recall still misses 4/16 critical patients; threshold reduces precision; no per-patient explanation without SHAP	Medium

3 Decision

Recommend Random Forest with ESI-1 probability threshold tuning for Phase 3, deployed as a passive decision support tool alongside triage nurse assessment.

4 Reasoning

- **Recall gain is clinically meaningful:** Random Forest + threshold raises ESI-1 Recall from 0.500 to 0.750, 12 of 16 critically ill patients correctly flagged vs 8 of 16. In a Caribbean ED with long reassessment intervals and documented understaffing (night shifts of 4 nurses against a target of 14, De Freitas et al., 2020), one additional critical patient caught is not a rounding error.
- **Interpretability is governable:** Random Forest feature importances provide a global audit trail without requiring SHAP tooling. Given the documented racial fairness concern from Week 5 (Black/African

American patients allocated to ESI 4 at 20.5% vs 12.8% for White/Caucasian patients), a model that can be interrogated at feature level is better positioned for the subgroup equity audit required before Ministry of Health deployment.

- **Compute cost is proportionate:** 35-second one-off training, 0.075 ms inference per patient. At Mercer General's patient volume this does not compound into an operational burden, directly addressing Martina Griffith's (Clinical IT Lead) documented concern about hidden compute costs from "just deploy it" decisions.

5 Things I Do Not Yet Know

- **Threshold transferability:** Whether the threshold that achieves 0.750 recall on Yale New Haven data will hold on Mercer General's patient presentations, higher prevalence of dengue, non-communicable disease burden, and walk-in patients, can only be answered with local validation data. Yang et al. (2024) demonstrated this gap is real and can produce highly variable performance in low/middle-income settings.
- **Clinical uptake in practice:** How Mercer General's triage nurses will respond to model alerts, whether they will trust, ignore, or appropriately integrate the recommendations, requires the pilot to begin before it can be assessed. This is why the human-in-the-loop design and weekly override-rate monitoring (Risk 7, automation bias, Week 4 risk register) are non-negotiable from day one.